P8139
Homework #3
Due 03/31/26

1. Read in the math/chem/bio scores data for PCA practice and answer the following questions.

```
mydata = read.table(file="~PCA_data.csv", header=TRUE, row.names=1, sep=",")
mydata.pca = prcomp(mydata, retx=TRUE, center=TRUE, scale=TRUE)

# set variable mean to zero and variance to one using "scale=TRUE"
# PCA scores for each sample are in mydata.pca$x
# loadings are in mydata.pca$rotation
# square roots of eigenvalues are in mydata.pca$sdev
# variable means are in mydata.pca$center
# variable standard deviations are in mydata.pca$scale

sd = mydata.pca$sdev
loadings = mydata.pca$rotation
rownames(loadings) = colnames(mydata)
scores = mydata.pca$x
plot(scores[,1:2],
xlim=c(min(scores[,1:2]),max(scores[,1:2])),ylim=c(min(scores[,1:2]),max(scores[,1:2])))
text(scores[,1], scores[,2], rownames(scores), col="blue", cex=0.7, pos=3)
screeplot(mydata.pca, type="lines")
```

a) Calculate PCA scores using loadings and original math/chem/bio scores and compare to output PCA scores from the R package prcomp.
b) Calculate percent variance explained of each component.
c) Plot i) PC1 vs PC2, ii) PC1 vs PC3, and iii) PC2 vs. PC3.

2. Read the paper “Transmission Test for Linkage Disequilibrium: The Insulin Gene Region and Insulin-dependent Diabetes Mellitus (IDDM)” by Spielman et al. published on American Journal of Human Genetics (AJHG) in 1993 and write one page summarizing what you learned.

   Other than your understanding of the test, in your summary, you can cite some sentences from the paper that clarify some of your understandings in concepts we have discussed in the class, such as linkage and linkage disequilibrium, samples used to examine these, etc.